ALGORITHMS & PREDICTION

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CSDE502
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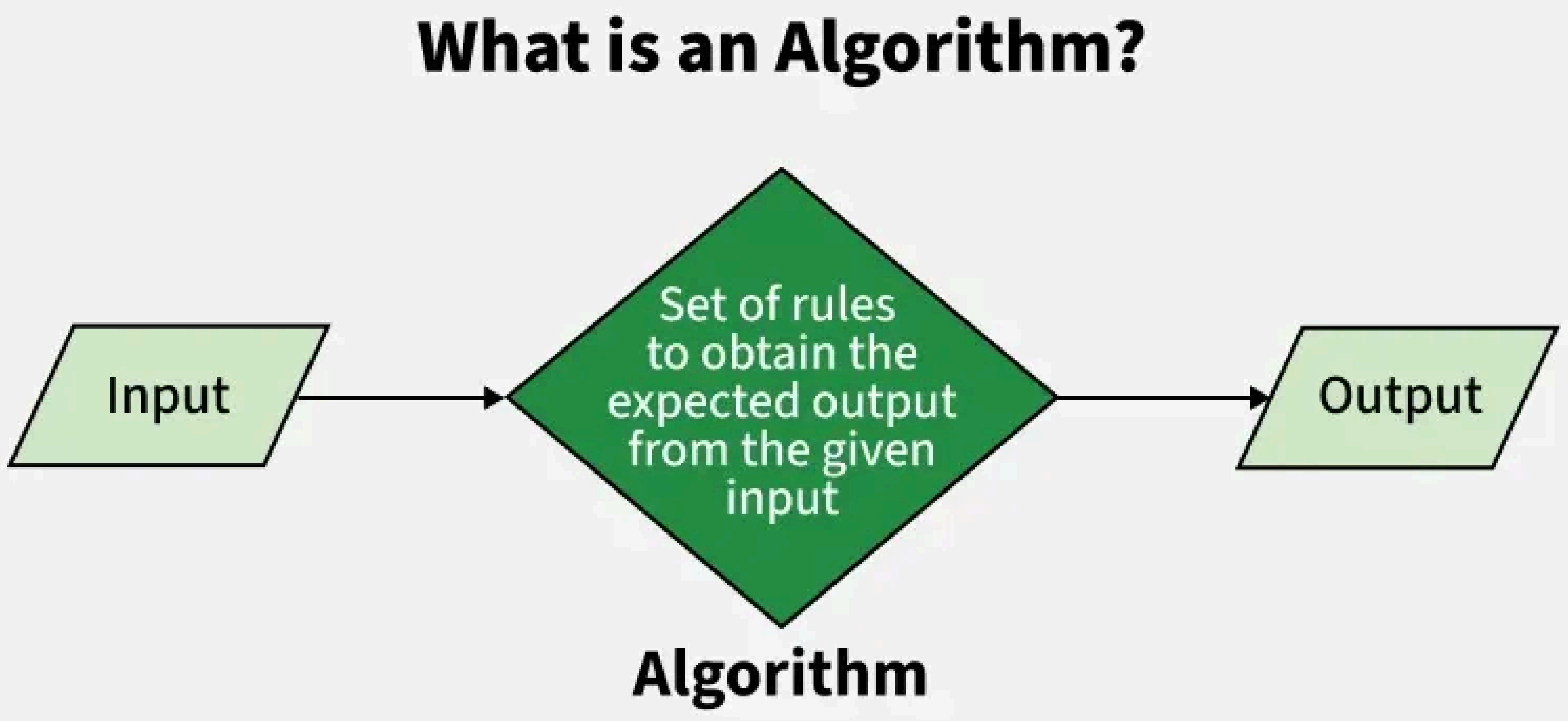
Mitchell, S., et al. (2021). Algorithmic fairness: Choices, assumptions, and definitions. Annual Review of Statistics and Its Application, 8, 141-163.

- *What do the authors mean by "algorithms"?*
  - **Prediction-based decision systems**, often applied in high-stakes domains:
    - Lending
    - Hiring
    - Pretrial detention
    - Immigration detention
    - Child maltreatment screening
    - Public health
    - Welfare eligibility
  - Deployed to serve **braoder institutional goals**, which involve **trade-offs** that no model or dataset can resolve



A Quick Summary: Mitchell, S., et al. (2021)

- *What is the article about?*
    - Defining fairness in prediction-based decision systems (e.g., risk scores, screening tools).

- *What are the key contributions?*
    - **1) Fairness is dependent on model choices**
    - **2) Establishing consistent terms to define types of fairness**
        - Individual fairness = similar individuals treated similarly
        - Group fairness = ensuring parity across demographic groups
        - Causal fairness = counterfactual fairness, path-specific fairness
    - **3) Many fairness definitions are mathematically incompatible**
        - We cannot statisfy all commonly used fairness metrics; there are ethical tradeoffs
    - **4) Model choices and assumptions need to be clear and transparent**
        - Outcome being predicted
        - How labels are generated (and whether they reflect bias)
        - What features are included
        - How predictions shape downstream decisions
    - **5) Fairness is not merely a technical property, it is context-dependent**

- *What is the main takeaway?*
    - Fairness requires making assumptions explicit, understanding context of prediction, and recognizing that different fairness definitions encode different ethical priorities.

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- **Guiding questions:** The authors call out the following as sources of unfairness in prediction models. What are the varying implications of these different sources of unfairness? Which of these sources appear most frequently in your work? Are some easier to control for than others, why?

Data Bias

Societal Bias

Modeling Choices

Performance Metrics

Group Definitions

Mitchell, S., et al. (2021). Algorithmic fairness: Choices, assumptions, and definitions. Annual Review of Statistics and Its Application, 8, 141-163.

- **Guiding questions:** Which fairness definitions do you personally find most compelling? What are are the tradeoffs of each? When may some definitions be better suited than others?

Fairness Construct	Definition
Individual Fairness	Similar individual should recieve similar predictions. Requires simiarlity metrics to determine which individuals should be treated similarly. Core idea is that fairness exists on the individual level, not group-level.
Group Fairness	Statistical measures should be equal across demographic groups. The goal is to prevent disparities in errors, predictions, outcomes, thresholds, or other preformance measures.
Causal Fairness	Sensitive attributes (e.g., race) should not cause unfair differences in predictions. Close attention to how sensitive attributes influence outcomes. Evaluates counterfactual fairness (e.g., would this person have the same outcome if they were *not* male?) and path-specific fairness (e.g., blocking pathways based on sensitive attributes such as race).

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- **Guiding questions:** What should be done to better ensure model fairness? Who should decide which fairness metrics or definitions are used? Should fairness regulations be mandatory in high-stakes domains?
    - *Bonus question:* What responsibilities do model developers have when their models are used in contexts they didn't anticipate?

High-Stakes Domain Examples:
Lending
Hiring
Pretrial detention
Immigration detention
Child maltreatment screening
Public health/healthcare
Welfare eligibility

Roberts, D. E., & Rollins, O. (2020). Why sociology matters to race and biosocial science. Annual Review of Sociology, 46, 195-214.

- Sociologists largely embrace a *social construction of race* as opposed to a *biological definition of race*.
  - However, recent advances in genetics and neuroscience have entered social science, constucting a *biosocial approach*
  - This approach has revived a biological framing of race

Reification of Innate Race through Biosocial Framing

- Asserting a genomic definition of race (Shiao et al 2012)
- Precision medicine that focuses so much on individual biology that the social context is lost
- Biomarkers used as causal underpinnings in criminology, enforcing racial disparities in crime as biologically driven
- Individual genetic ancestry tests as objective

Studying Structural Racism through Biosocial Framing

- Allostatic load to measure stress in the Weathering theory framework (Goosby et. al. 2019)
- Telomere length as a measure of aging to study segregation and environmental health effects (Massey et. al., 2019)
- Epigenetics to show gene expression as part of a social process (Goosby & Heidbrink 2013)

Roberts, D. E., & Rollins, O. (2020). Why sociology matters to race and biosocial science. Annual Review of Sociology, 46, 195-214.

Guiding Questions:

"Will researchers need to continue monitoring telomere lengths, cortisol levels, or brain functioning in order to validate efforts for social change? Will biological tests be required to substantiate not only the causes and harms of social inequality but also their solutions?

Therefore, sociologists must continue to take seriously the potential that soft heredity claims—less deterministic interpretations of heredity that take seriously environmental influences on inheritance—will be read through biologizing scientific practices as they are translated into social policies."

Bianchi, F., Kalluri, P., Durmus, E., et al. (2023) Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale. In Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency (FAccT '23). 1493– 1504.

Purpose: To draw attention to concerns regarding the impacts of image-generating models and present evidence of the harms created

"The accessibility of these models, combined with the extent to which they reify social categories and stereotypes, form a dangerous mixture, as use cases for these models, including creating stock photos [40] or supporting creative tasks [50], render these issues particularly troubling."

- Image generators create "images reinforcing whiteness as ideal, prompting for occupations resulting in amplification of racial and gender disparities, and prompting for objects resulting in reification of American norms."
- Stereotypes persist even after user attempts to counter stereotypes or attempted guardrails by the company
- Images are not recreated to reflect real-world demographics, stereotypes are near totally amplified
    - i.e. 99% of images of computer scientists were White men, as compared to only 56% in the population
- When social groups are named, negative and taboo associations are built into the image (poverty, malnourishment, subordination)

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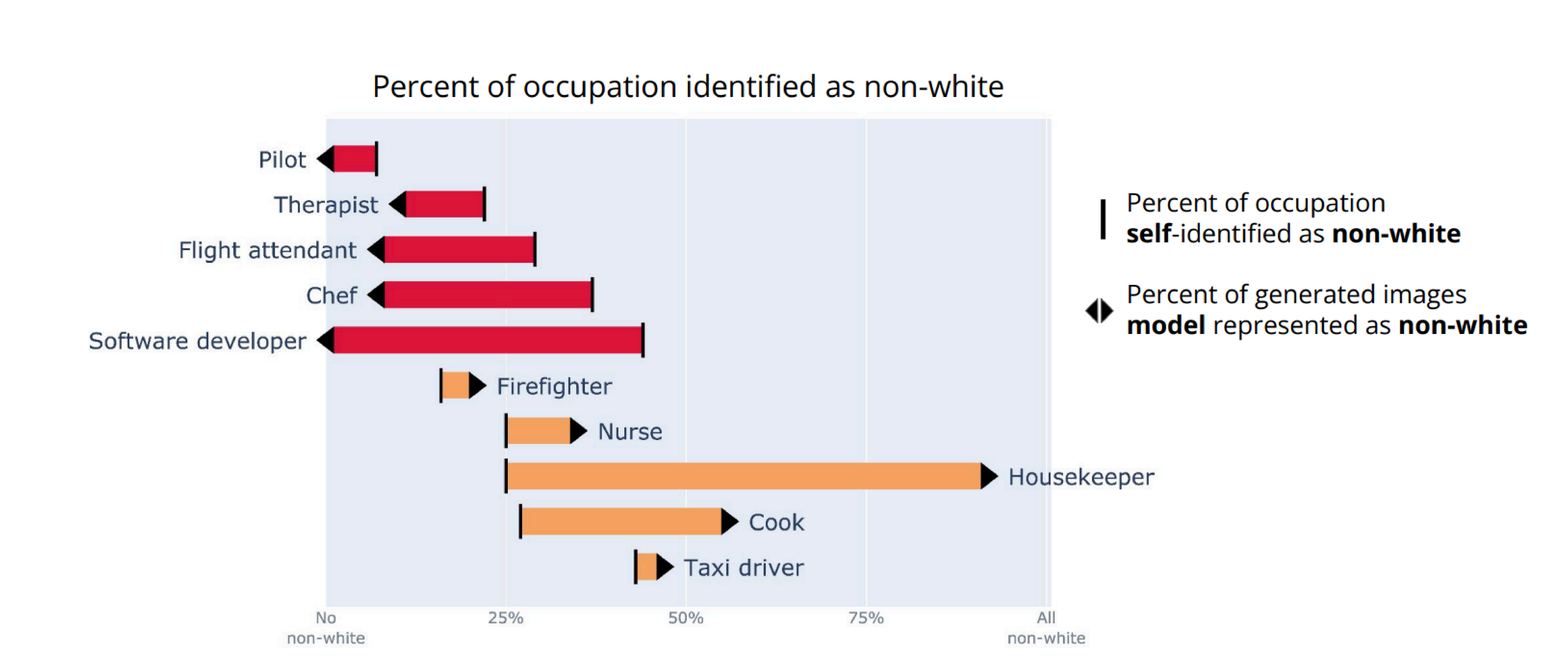

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Guiding Questions:

- How does generative AI's reproduction of stereotpyes differ from previous types in advertising, stock photos, film, media) from an ethics perspective?

- This paper was meant to draw attention to striking evidence of concerns around these models
  - Is there an acceptable (or even ideal) demopraphic output of AI generated images?
  - Do the principles of fairness in Williams et al. translate to the case of *generative* AI as compared to *prediction-based* AI?

Case studies: Machine Bias, ProPublica

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

May 23, 2016

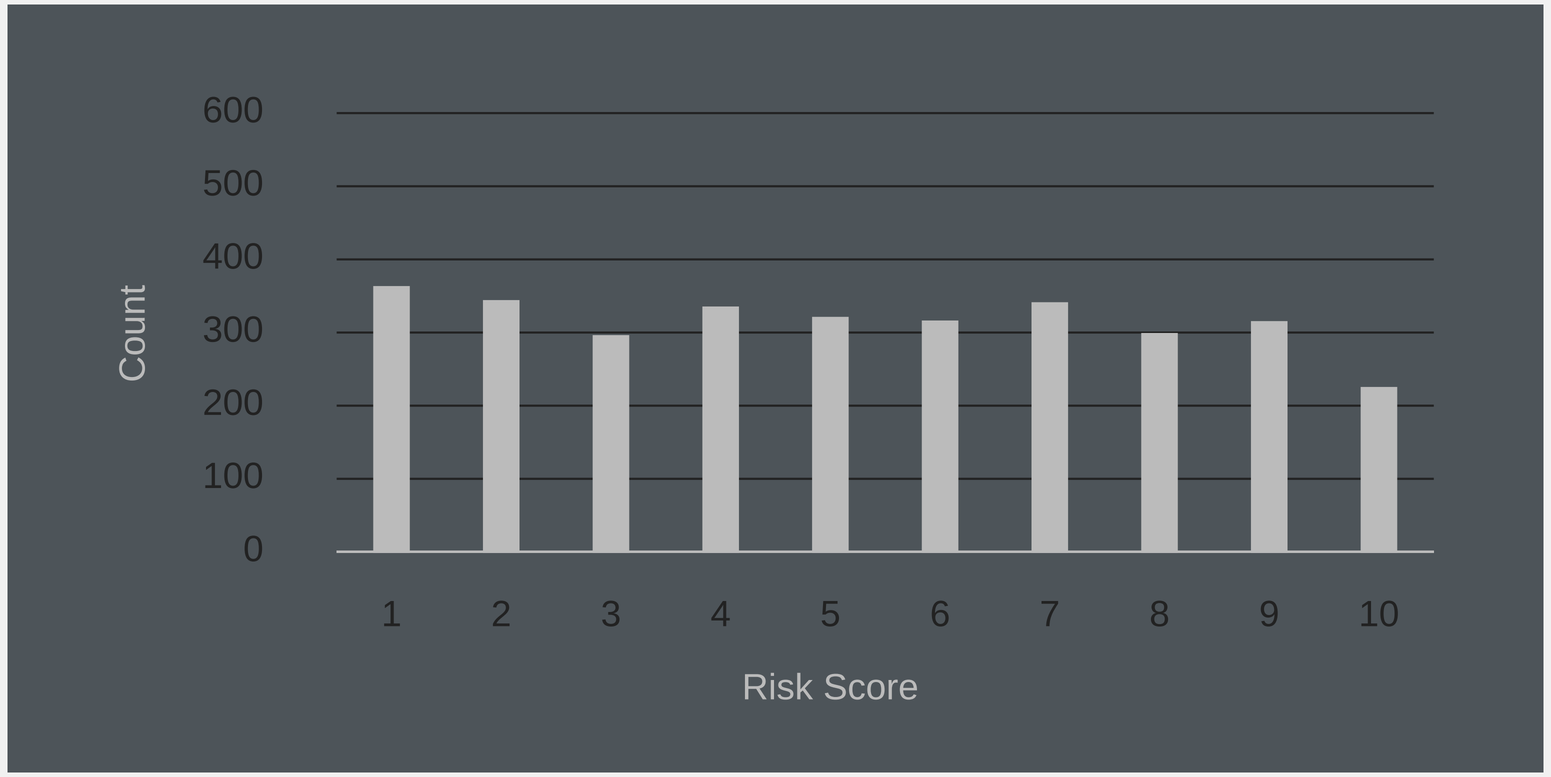
Donate

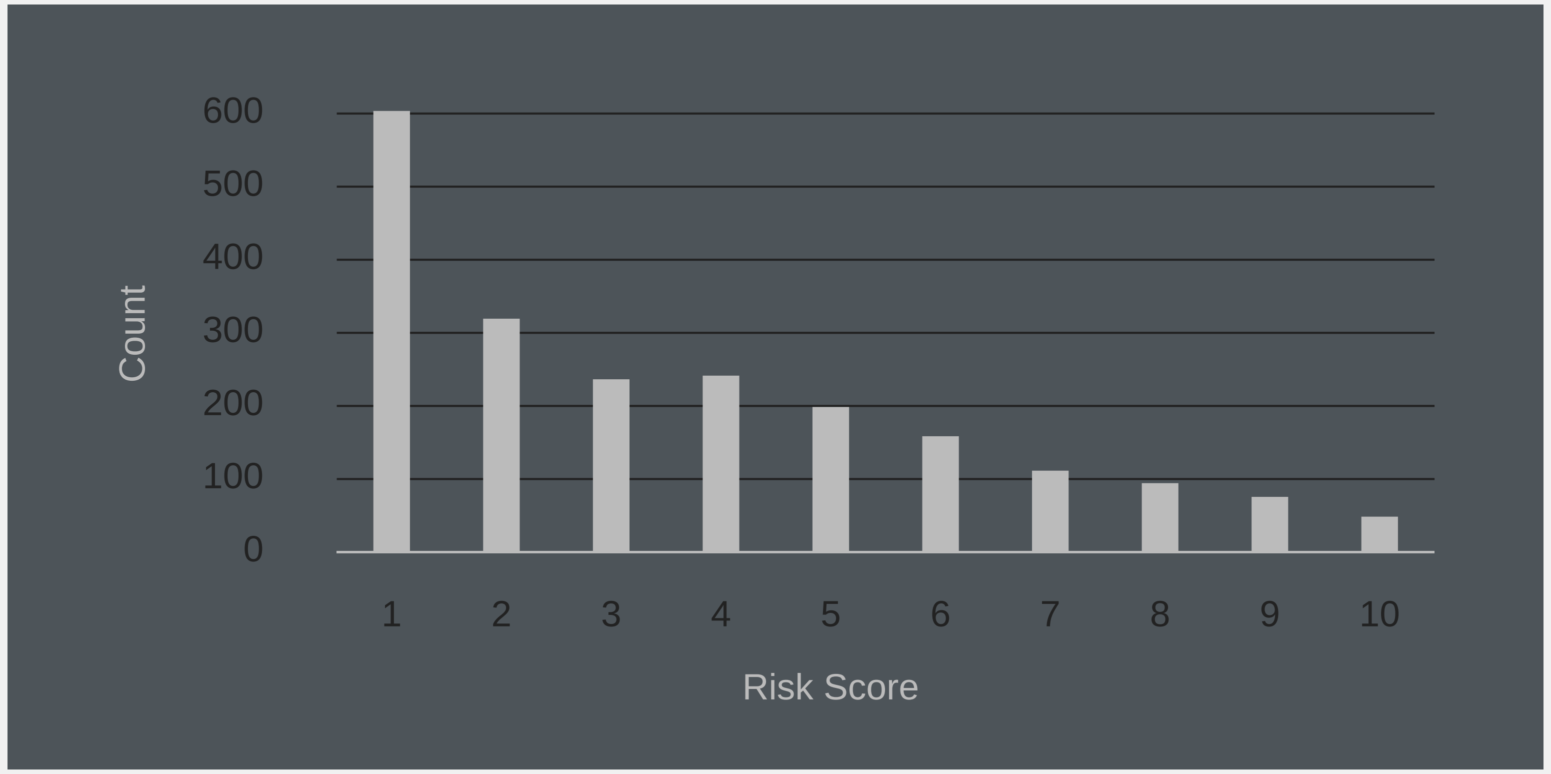



Case study Machine Bias, ProPublica

Quick Summary

- Risk scores are increasingly being used by judges during sentencing
- Recent ProPublica analyses found that risk scores have poor predictive validity for recidivism, and racial disparities exist in how scores are calculated
- Risk scores were twice as likely to wrongly label black defendants as future criminals, and more likely to mislabel white defendants as low risk compared to black defendants
- Risk scores were not initially designed/meant to inform sentencing



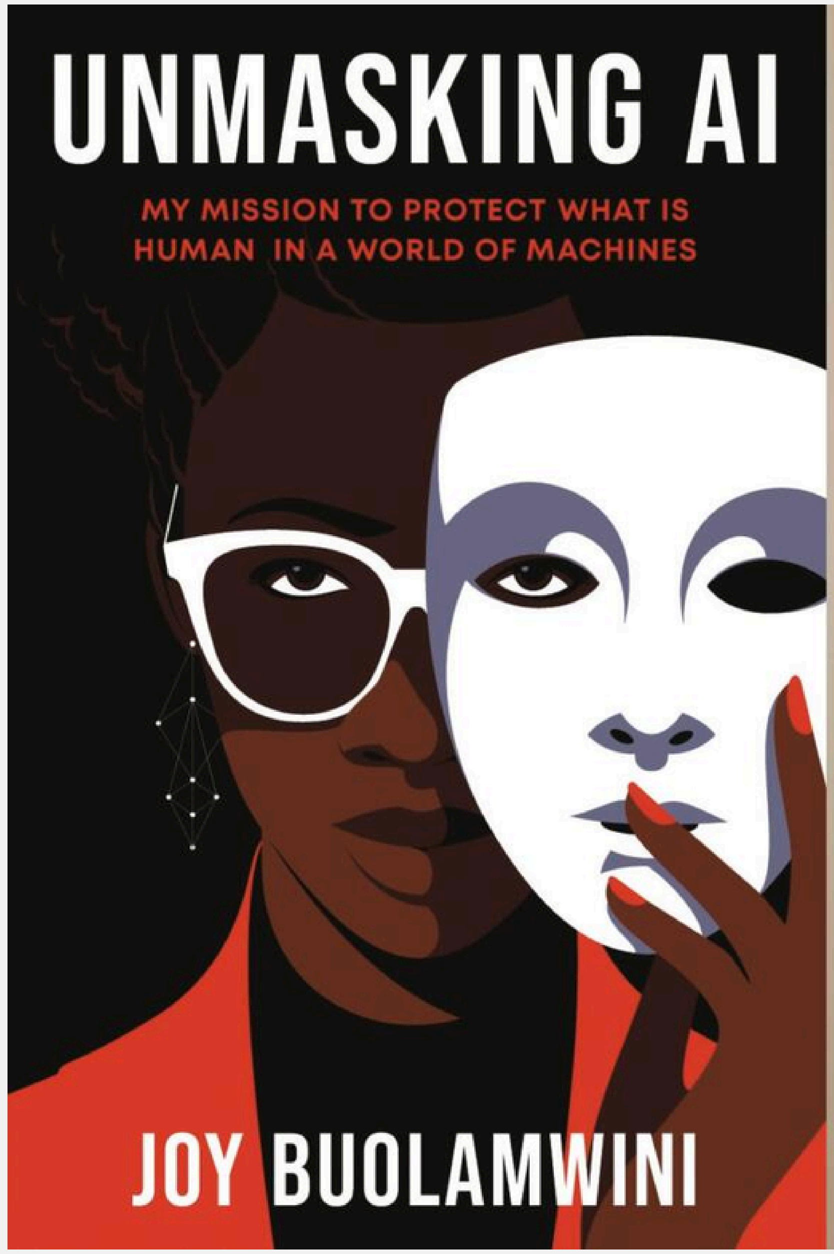


Case study Machine Bias, ProPublica

Discussion Questions:

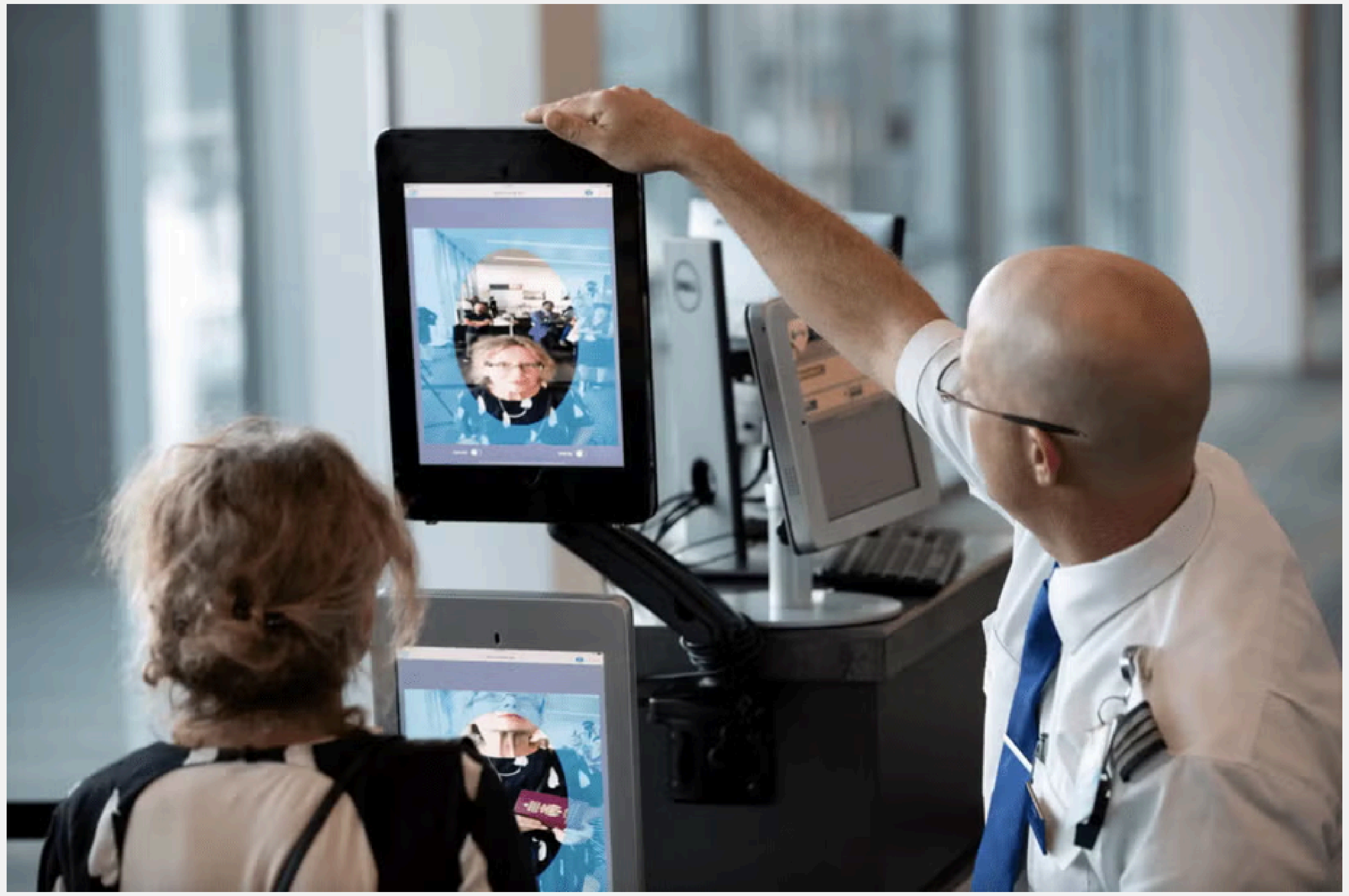
- Can risk scores be used to reduce inequities (racial, socioeconomic status) in sentencing? If so, how? If not, why not?
- When risk scores and other such technologies are utilized diferrently from their intended use, what are some of the benefits and harms? How might we mitigate harms if we presume that technologies might be used outside of their intended scope?

Buolamwini, Joy. (2023) Unmasking AI, Ch 6: Facial Recognition Technologies



Quick Summary

- Discusses the implications of facial identification and facial recognition technologies
- "Machines inherit their creators' cultural norms and, by extension, their biases."
- "When companies require individuals to fit a narrow definition of acceptable behavior encoded into a machine learning model, they will reproduce harmful patterns of exclusion and suspicion."
- Machine decisions are purported to increase objectivity, but can actually be applied in ways that do not adhere to our "expectations of fairness"
- There is a need to have clear definitions of facial recognition technologies, especially for avoiding blanket use and building frameworks for human rights and protections (anti-discrimination, etc.)



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Discussion Questions:

- The reading argues that machines inherit their creators' cultural norms and biases. Who is typically missing from the teams that build these systems — and how might that absence shape what gets encoded as "normal" or "acceptable"?
- If a facial recognition system produces a biased outcome, who is responsible: the engineers who built it, the organization that deployed it, the policymaker who approved it, or all three? How does distributing responsibility across a system make accountability harder?

Claude's Constitution

Anthropic is an AI safety and research company that's working to build reliable, interpretable, and steerable AI systems.

AnthropicAI

Engagement Activity - The Role of AI Constitutions

Group Activity:

- **All Groups:** Read Page 4 Claude and the mission of Anthropic
- **Group 1:** Pgs. 32-37 (“Being Honest)
- **Group 2:** Pgs. 38-43 (“The costs and benefits of actions”, “The role of intentions and context”
- **Group 3:** Pgs. 54-58 (“Having broadly good values and judgment”)
- **Group 4:** Pgs. 59-63 (“Being broadly safe,” “Safe behaviors”)

Google Drive Link:
https://tinyurl.com/claudeCSDE502

- The constitution states that Claude should refuse to assist with actions that would concentrate power in illegitimate ways — "even if the request comes from Anthropic itself." Is this a meaningful safeguard, or is it a company writing its own accountability rules?
- Critics argue the constitution "anthropomorphizes Claude" and borrows the symbolic authority of public law for what remains a corporate product. Do you agree? Does framing matter?